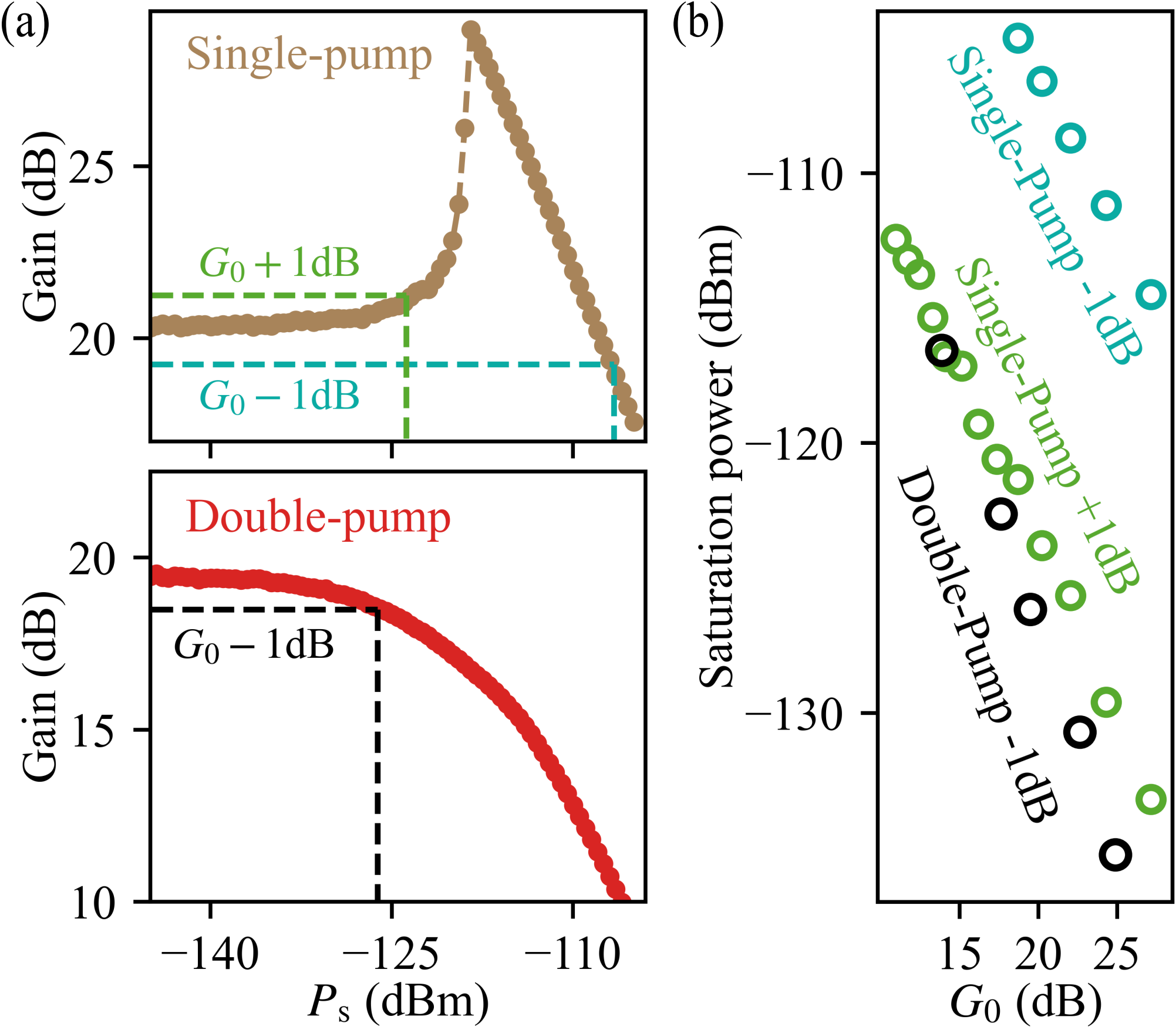
(a)
Single-pump
$G_0 + 1$dB
$G_0 - 1$dB
Gain (dB)
25
20
Double-pump
$G_0 - 1$dB
Gain (dB)
20
15
10
−140
−125
−110
P_s (dBm)
(b)
Saturation power (dBm)
−110
−120
−130
Single-Pump -1dB
Single-Pump +1dB
Double-Pump -1dB
15
20
25
G_0 (dB)